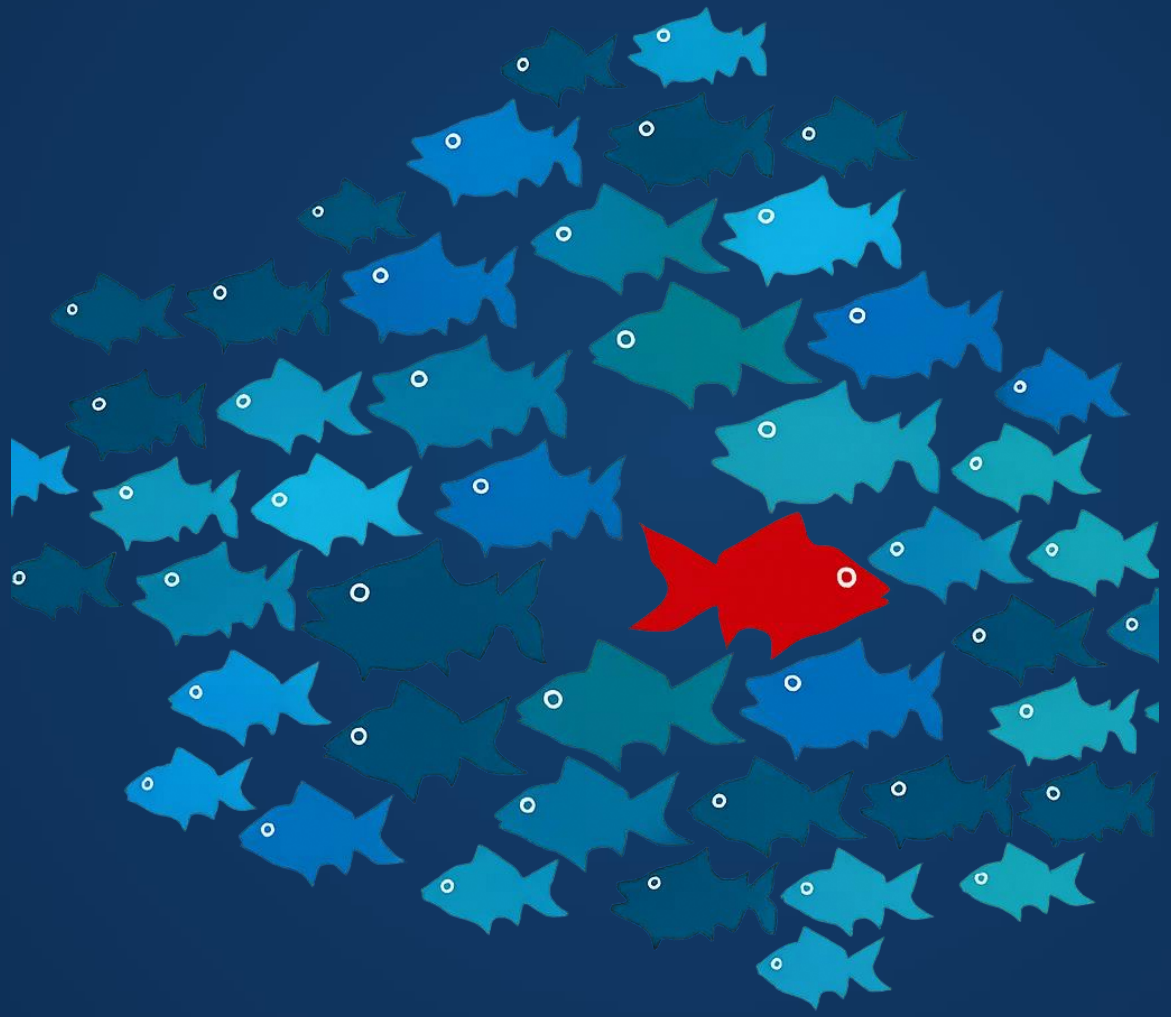
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CS5702 MODERN DATA · LECTURE 4

Exploratory Data Analysis & Visualisation

Spot the red fish before you build the model.



Dr Ferdoos Hossein Nezhad · Department of Computer Science

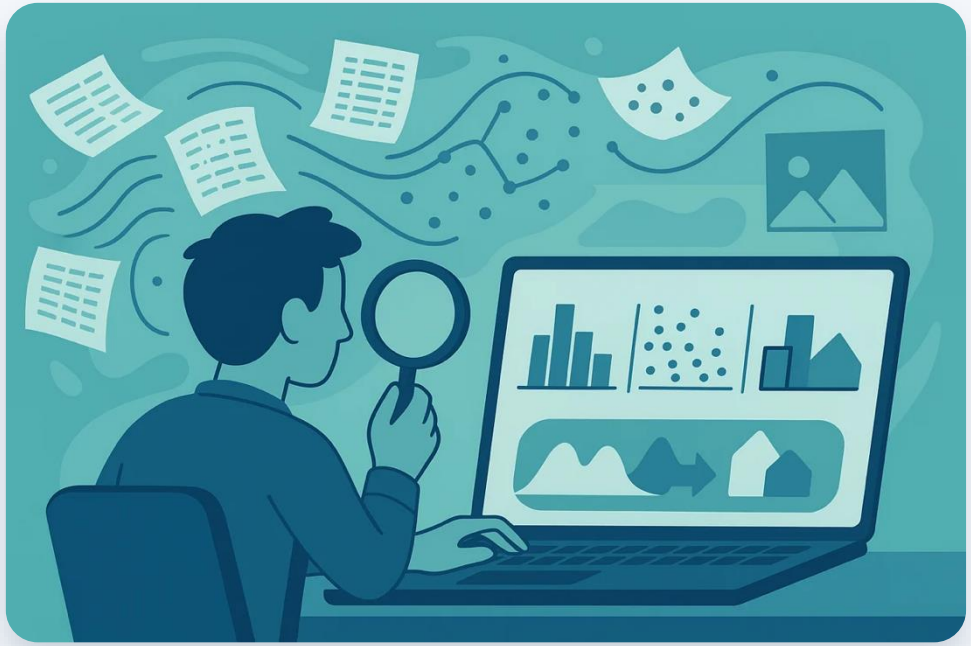
TODAY'S PLAN

"Find the hidden story."

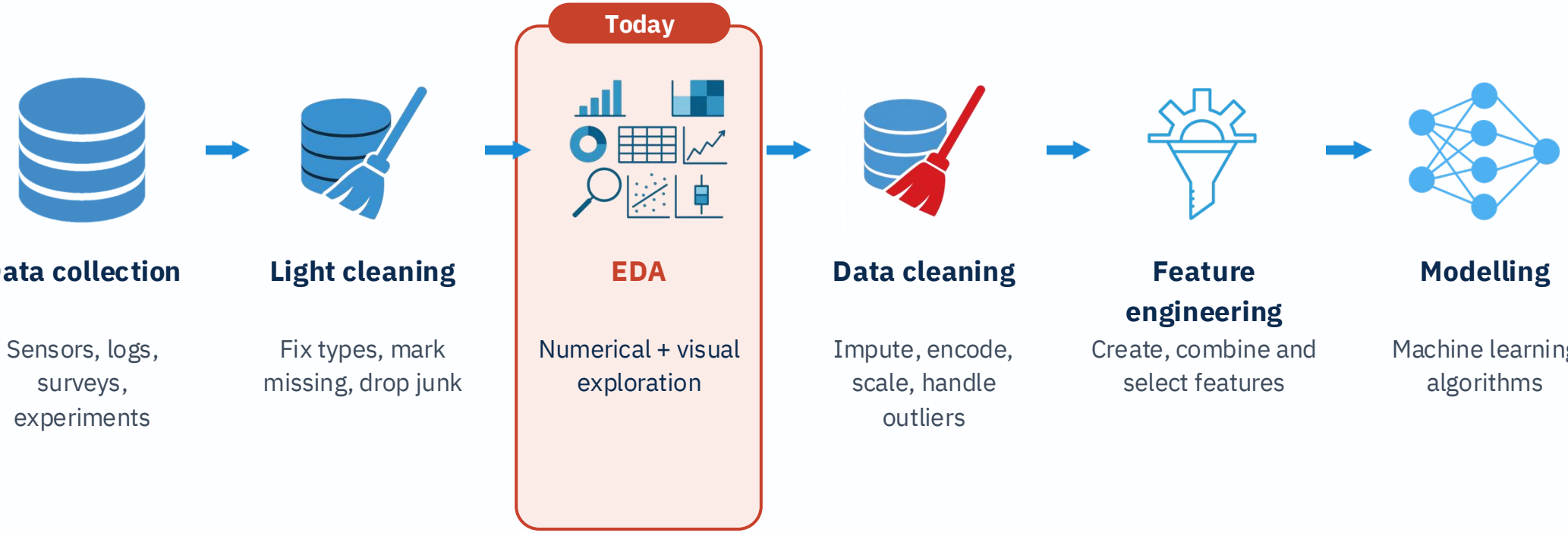
1 **What is EDA?**
Understand the data before building any model

2 **Patterns and variation**
Relationships, trends and surprises under the surface

3 **Data visualisation in R**
Histograms, scatterplots, boxplots and ggplot2



The data journey



What we build depends on how carefully we understand and prepare the data.

STEP 2 OF THE JOURNEY

Light cleaning: tidy up before you explore

Fix data types

"12.3" → numeric

"2023-05-01" → date

Mark categorical variables as factors

Standardise missing values

"NA", "?", -99 → NA

Don't impute yet — just mark them

Drop meaningless columns

Pure ID columns

Completely empty columns

Goal: a consistent, interpretable dataset — **ready for EDA**.

STEP 3 OF THE JOURNEY

Two sides of EDA

Numerical EDA

Summarise with numbers — and the R to do it

Shape of the dataset	dim()
Data types	str()
Unique values	unique()
Basic statistics	summary()
Statistical tests	t.test()

Visual EDA

See the patterns — from one variable to many

Univariate · one variable
Histogram, boxplot, KDE, violin plot

Bivariate · two variables
Scatterplot, boxplot by category

Multivariate · many variables
Heatmap, scatterplot matrix, clustered heatmap

WHY IS EDA USEFUL?

A real study: does the model fit the data?

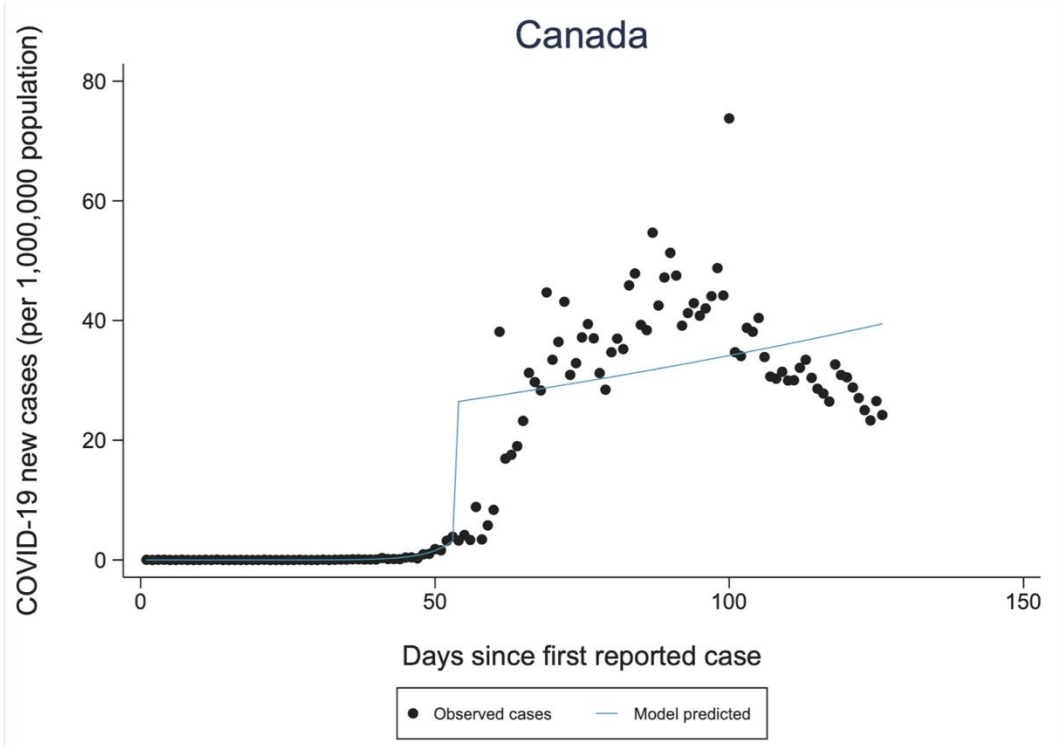
Islam et al. (2020), BMJ 370

149 countries: government interventions vs **daily new COVID-19 cases**

Interrupted time series with **regression discontinuity** modelling

A complex method — so for now, let's just **eyeball the fit**

> *Does the blue line really capture the dots?*

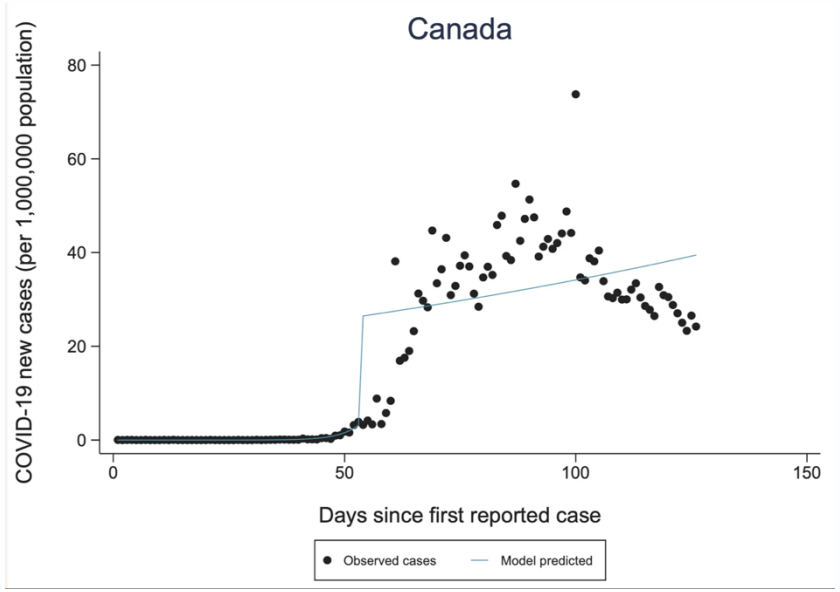


Canada · dots = observed cases · line = model prediction

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WHY IS EDA USEFUL?

Five lessons from discontinuity modelling



Islam, N. et al. (2020) Physical distancing interventions and incidence of coronavirus disease 2019: natural experiment in 149 countries. *BMJ*, 370, m2743.

The Canada chart is **buried deep in the appendix**

The fit looks **better for many other countries** — but not all

The authors are **expert epidemiologists** — experts can still miss things

Don't be **intimidated** by names like the "Schneidelgrüber Hyper-Transverse Spline Method"

Always start by looking at the data — it spots problems early



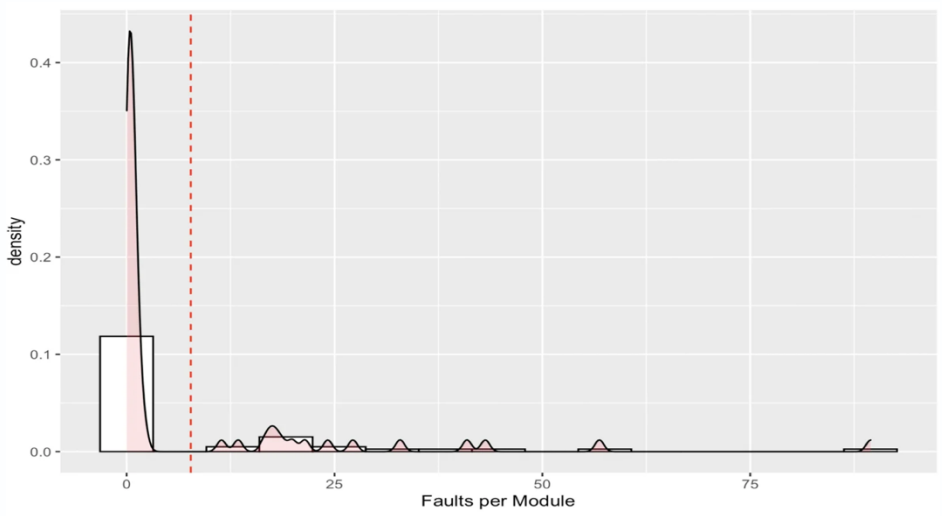
Never assume a model is right. Start with your own eyes.

Explore the data, challenge assumptions — and don't let the jargon fool you.

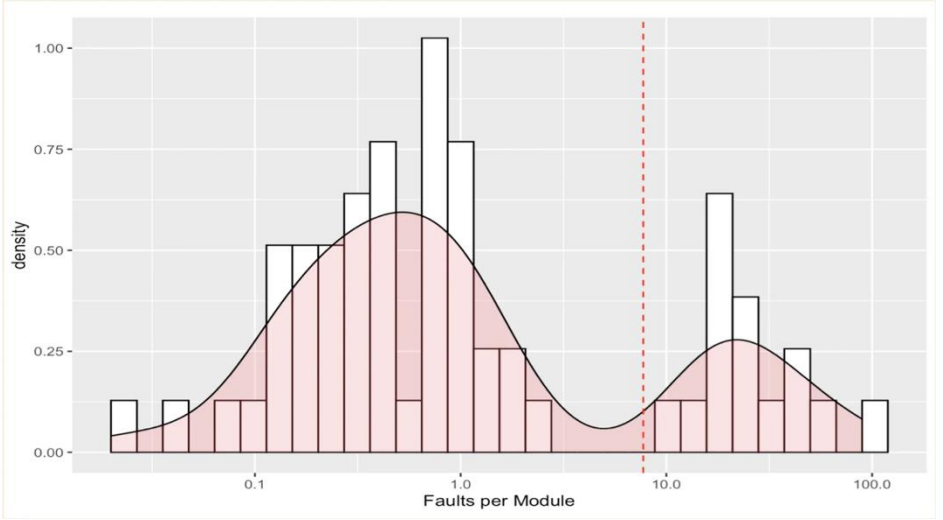
WHY EYEBALLING THE DATA MATTERS

Change the view, see the detail

Before · **linear axis: points pile up near zero**



After · **log axis: hidden structure appears**



No rule says "use a log scale". The choice comes from **curiosity, judgement and practice**.

HOW OUR VIEW OF EDA EVOLVED

From procedure to philosophy

1970S · THE FATHER OF EDA

John Tukey

"Procedures for analysing data, techniques for interpreting results, and methods for planning data collection."

- Statistical procedures
- Accuracy and precision
- A systematic process
- Interpretation and validation

TODAY · MODERN PERSPECTIVE

The NIST view

"An approach — mostly graphical — to understanding data."

- Maximise insight into a dataset
- Uncover structure; extract key variables
- Detect outliers and anomalies
- Test assumptions; build simple models

Tukey gave us **discipline** · NIST added **discovery** → be **rigorous and curious**

WHY WE DO IT

Five goals of EDA

Understand your data

What each variable means, its values and gaps

Discover questions

Patterns and surprises reshape what we ask

Assess data quality

Outliers, inconsistencies, recording errors

Summarise key features

Plots and statistics tell the story

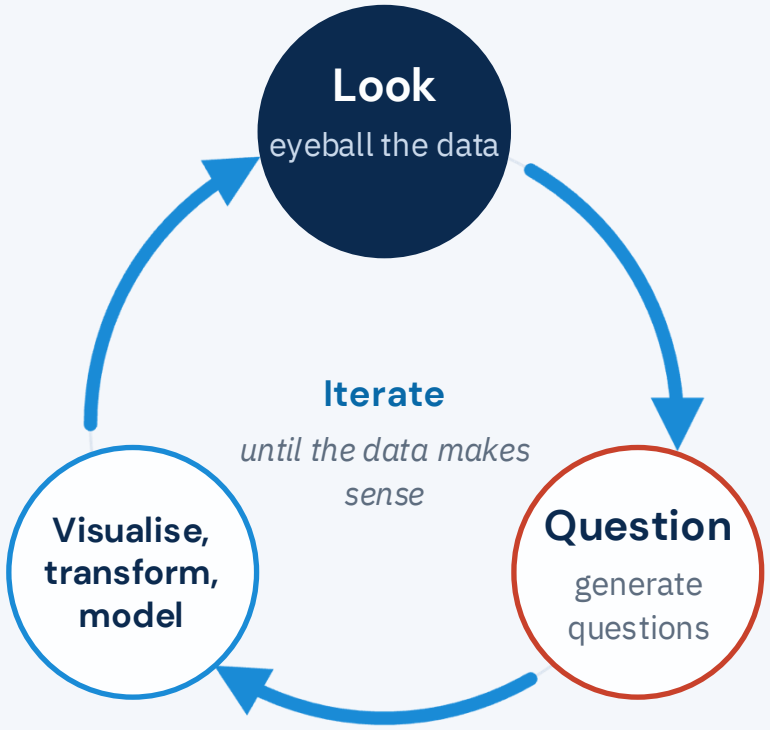
Guide next steps

Informed cleaning, transforming and modelling

See before we model · **question** before we assume · **understand** before we predict

MAIN STEPS IN EDA

An iterative cycle, not a checklist



> *Like a detective at a scene: the data is your evidence. Each clue changes where you look next.*

Three guiding questions

? What **data types** am I working with?

? What **variation** exists within my variables?

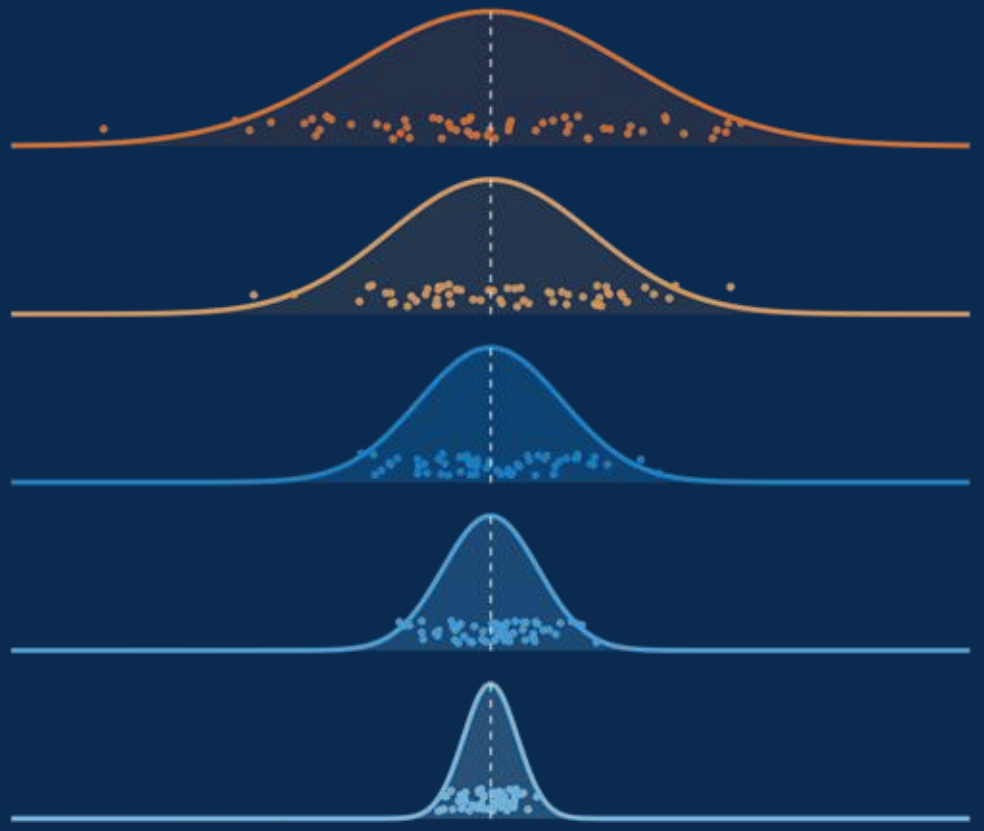
? What **covariation** exists between my variables?

EDA is not a well-defined process — but **aim to be systematic**.

2

Patterns and variation

The EDA toolkit, a worked example in R, and why variance matters



EDA TOOLS

Your detective's toolkit

Data type	Tool	What it tells you
Numeric	Boxplot	Spread and outliers at a glance; compare groups
Numeric	Histogram	Shape: clustered, skewed or evenly spread?
Numeric	Interquartile range	How concentrated the middle 50% is; flags outliers
Numeric	Q–Q plot	Is it normal? Points drift off the line if not
Categorical	Counts & bar charts	Which categories dominate; how groups compare
Categorical	Contingency tables	How two categorical variables relate (e.g. location × response)

One purpose: **make invisible patterns visible.**

Eyeballing the USArrests data in R

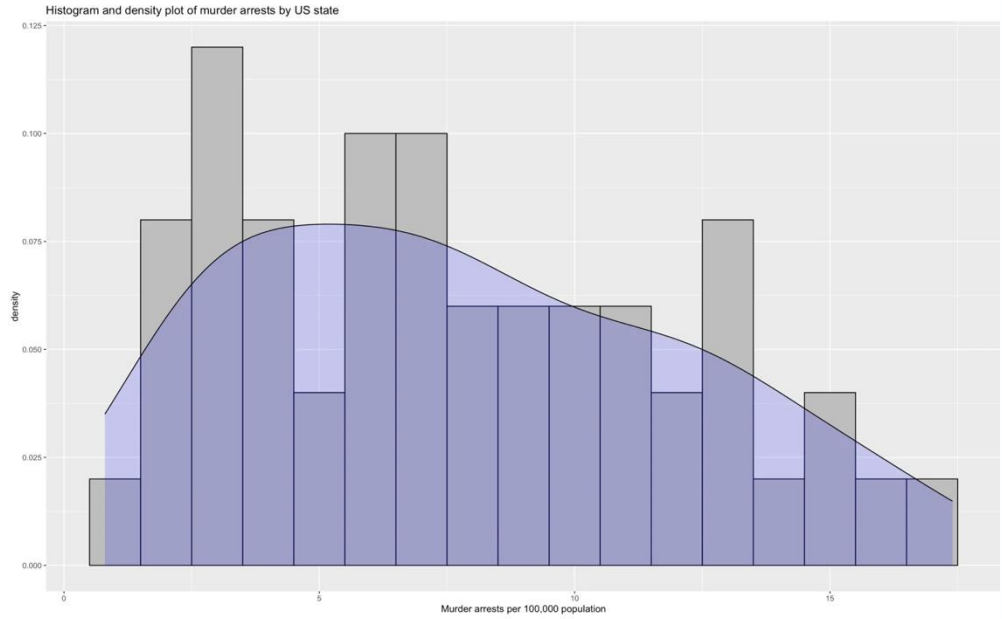
1 · Load and peek

```
> data("USArrests")
> head(USArrests)
            Murder Assault UrbanPop Rape
Alabama       13.2     236       58 21.2
Alaska        10.0     263       48 44.5
Arizona        8.1     294       80 31.0
Arkansas       8.8     190       50 19.5
California     9.0     276       91 40.6
Colorado       7.9     204       78 38.7
```

2 · Summarise

```
> summary(USArrests)
     Murder          Assault         UrbanPop          Rape       
 Min.   : 0.800   Min.   : 45.0   Min.   :32.00   Min.   : 7.30  
 1st Qu.: 4.075   1st Qu.:109.0   1st Qu.:54.50   1st Qu.:15.07  
 Median : 7.250   Median :159.0   Median :66.00   Median :20.10  
 Mean   : 7.788   Mean   :170.8   Mean   :65.54   Mean   :21.23  
 3rd Qu.:11.250   3rd Qu.:249.0   3rd Qu.:77.75   3rd Qu.:26.18  
 Max.   :17.400   Max.   :337.0   Max.   :91.00   Max.   :46.00  
```

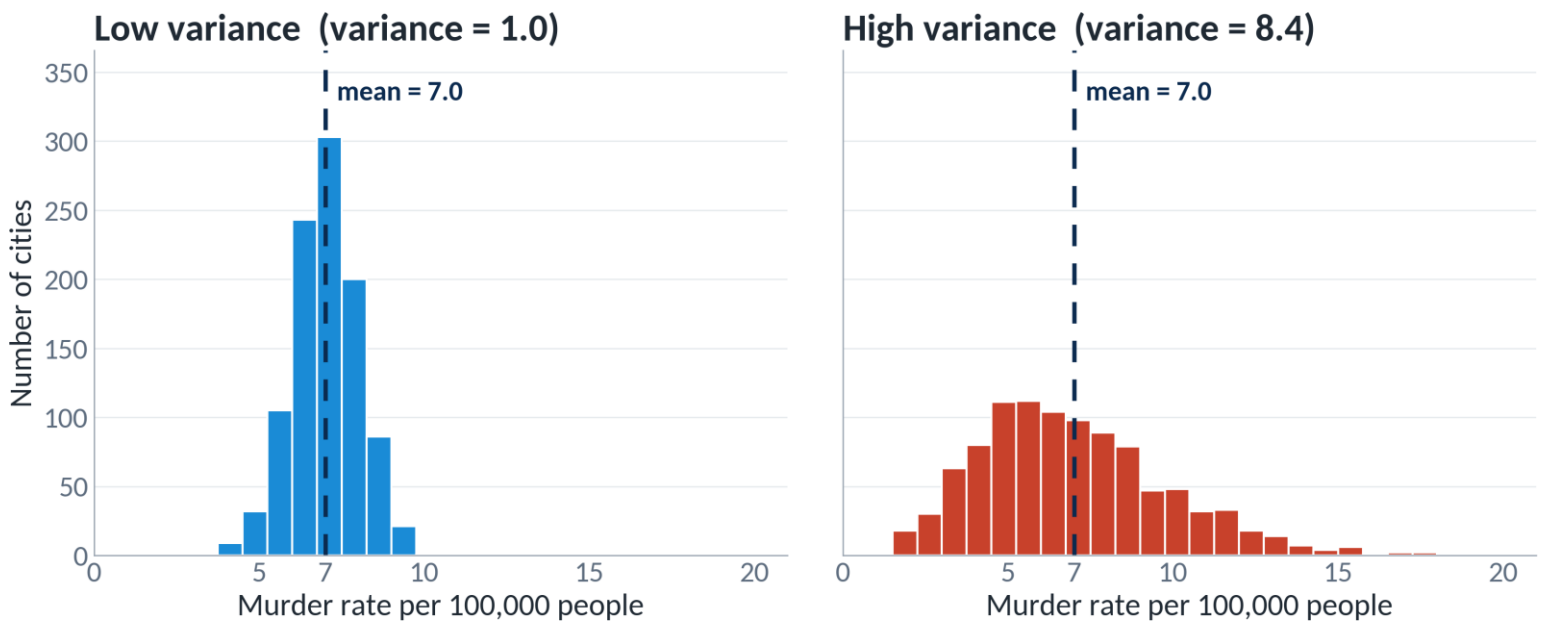
3 · Plot the distribution



What shape do you see? Skew? Peaks? Gaps?

PATTERNS AND VARIATION

Variance matters: same mean, different story



7.0

mean murder rate in **both** groups

1.0

variance: cities are consistent

8.4

variance: some safe, some dangerous

The mean alone can mislead — **always check the spread**.

3

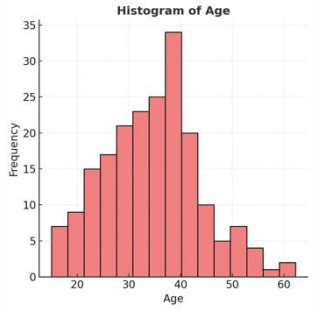
Visualising data in R

One variable, two variables, many variables — then ggplot2

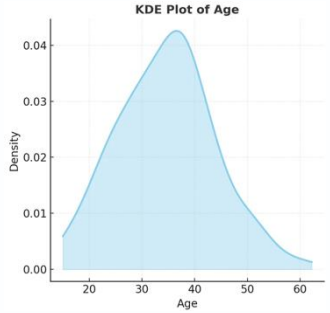


VISUAL EDA · UNIVARIATE

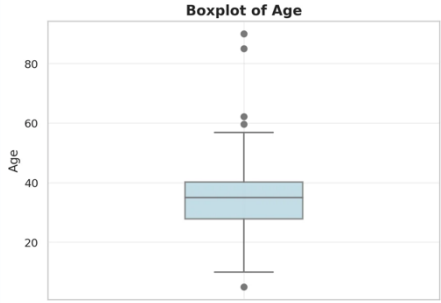
One variable at a time



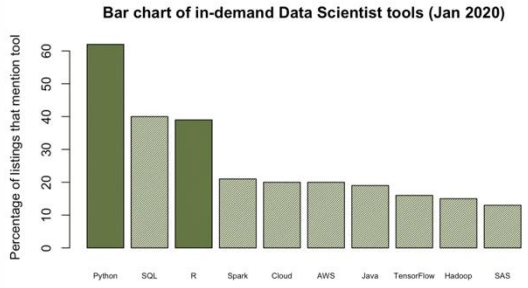
Histogram · shape, skew, peaks



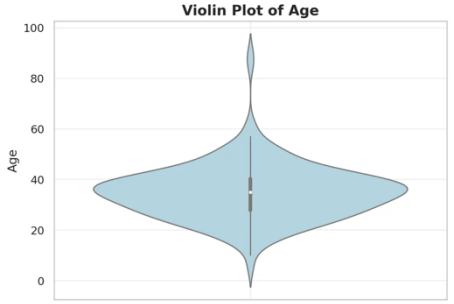
KDE plot · a smooth histogram



Boxplot · median, IQR, outliers



Bar chart · counts for categories



Violin plot · boxplot + density

RULE OF THUMB

Numeric → histogram, KDE, boxplot, violin

Categorical → bar chart

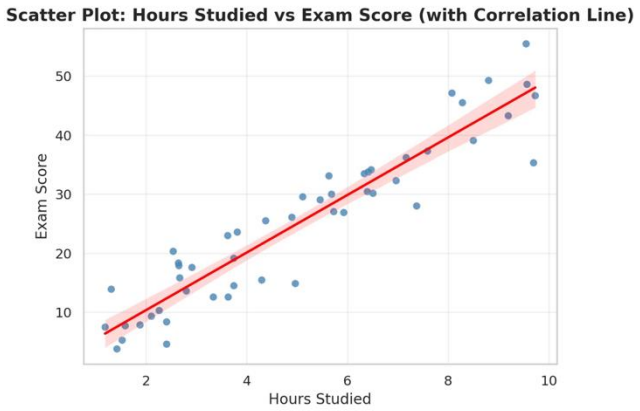
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VISUAL EDA · BIVARIATE

How do two variables relate?

Numeric × numeric

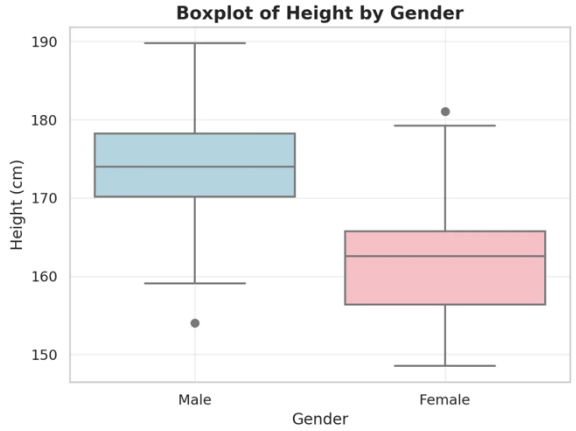
Scatter plot



More study hours → higher scores: a **positive trend**

Numeric × categorical

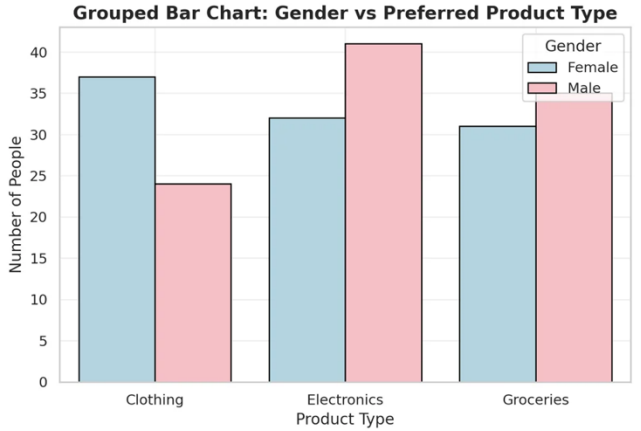
Boxplot by category



Height by gender: compare **centre, spread, outliers**

Categorical × categorical

Grouped bar chart

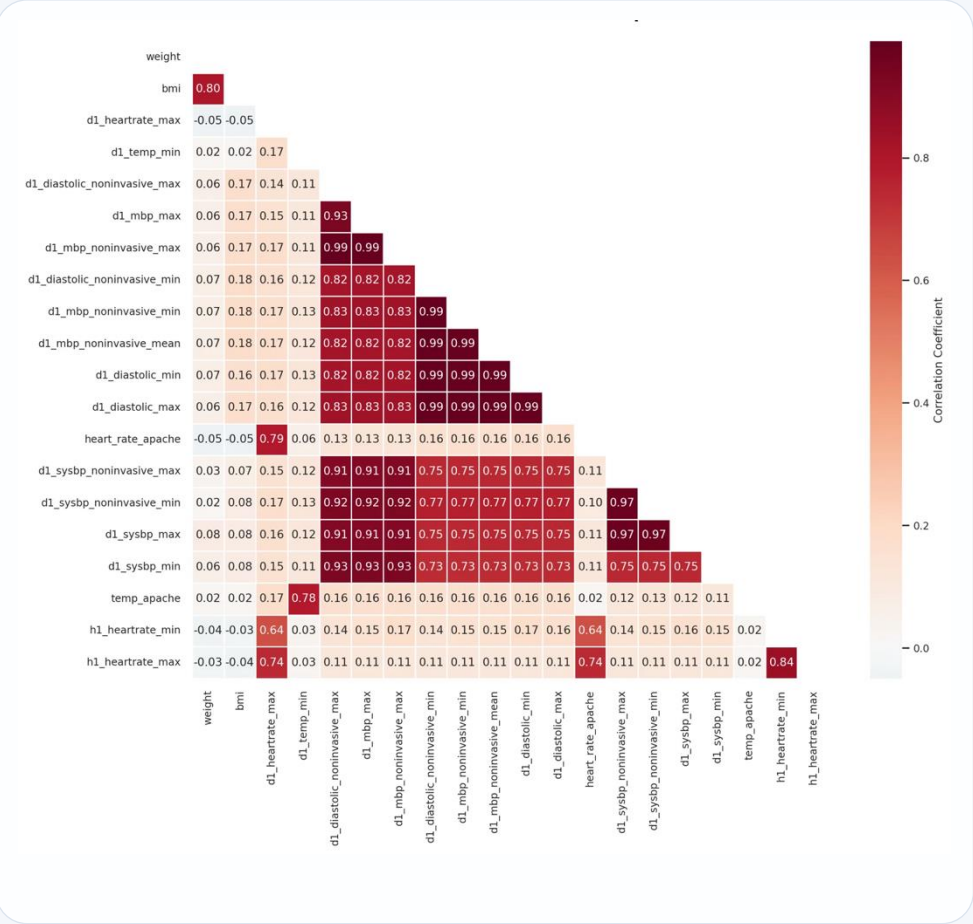


Product preference by gender: **compare groups**

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VISUAL EDA · MULTIVARIATE

The correlation heatmap



How to read it

Dark red · strong positive correlation

Pale · weak or no correlation

Blue (in other palettes) · negative

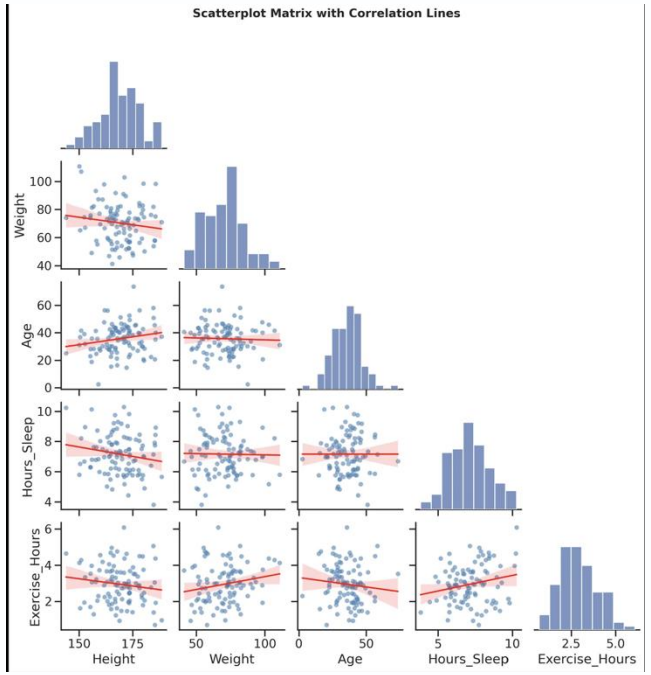
Use it to

- Spot **redundant** variables to drop
- Find **meaningful relationships** worth testing
- Reduce **multicollinearity** before modelling

VISUAL EDA · MULTIVARIATE

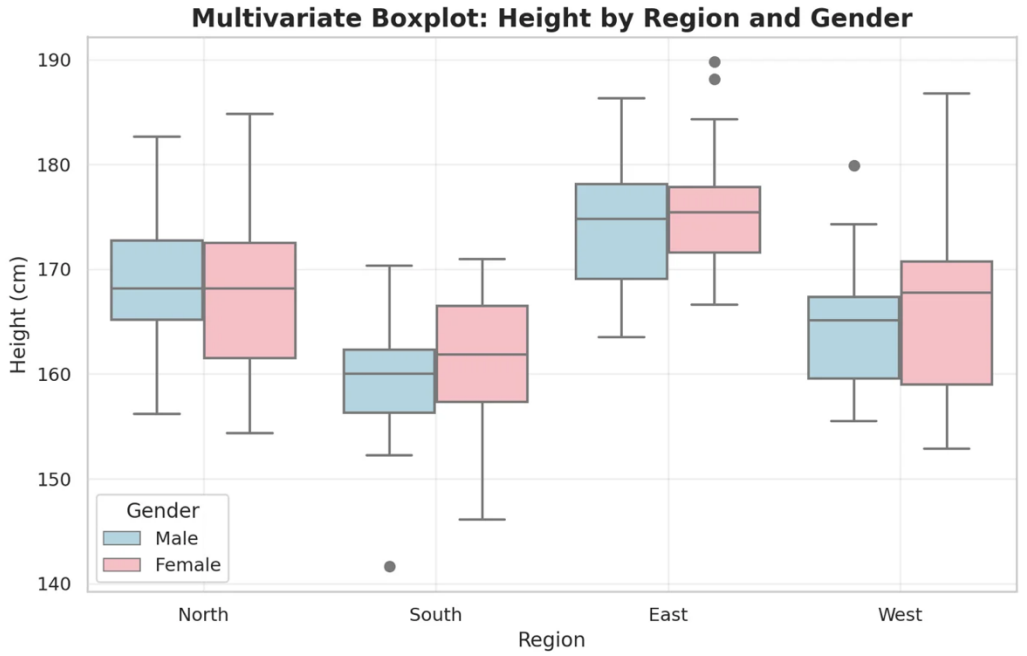
Every pair at once, and groups within groups

Scatterplot matrix



Diagonal = each variable's distribution; off-diagonal = pairwise trends

Multivariate boxplot



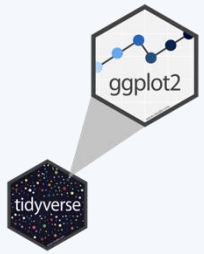
Height by **region** and **gender**: patterns that depend on more than one factor (interactions)

CHEAT SHEET

Which plot should I use?

My question	Variables	Try
What shape is this variable?	1 numeric	Histogram, KDE, violin
How spread out? Any outliers?	1 numeric	Boxplot
How often does each category occur?	1 categorical	Bar chart
Do two numbers move together?	2 numeric	Scatter plot
Do groups differ?	numeric + category	Boxplot by category
Do two categories relate?	2 categorical	Grouped bars, contingency table
How do many variables relate?	many	Heatmap, scatterplot matrix

Welcome to {ggplot2}



Grammar of Graphics (Wilkinson) — hence "gg"

Layers: add detail step by step

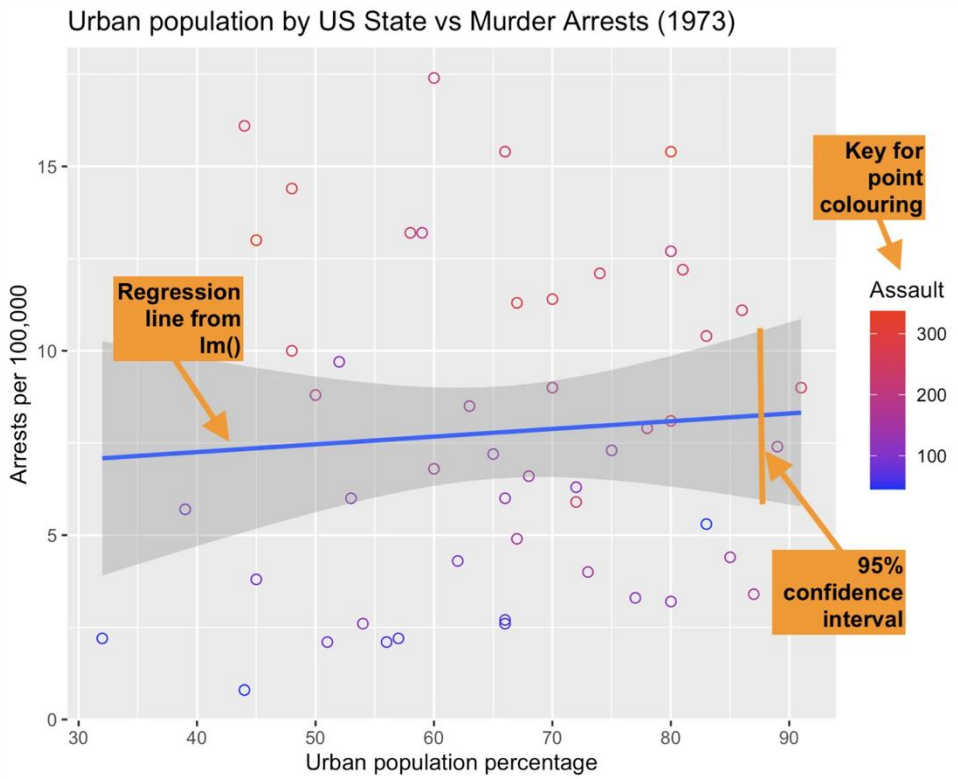
Colour and facets for many variables

Themes and nice defaults (usually!)

```
install.packages("tidyverse") # once
library(ggplot2) # every new session

ggplot(USArrests, aes(UrbanPop, Murder)) +
geom_point(aes(colour = Assault)) +
geom_smooth(method = "lm")
```

Part of the tidyverse · well documented on tidyverse.org · cheatsheet available



Points + colour key + regression line + 95% confidence band

WEEK 4 LEARNING OUTCOMES

By the end of this session, you can…

- **Explain** the purpose and nature of exploratory data analysis
- **Apply** simple techniques in R to describe and understand datasets
- **Generate and interpret** basic plots to visualise different variables
- **Use and adapt** ggplot2 examples to create customised visualisations

KEEP EXPLORING

Further reading

R for Data Science

Grolemund & Wickham (2018)

Chapter 7

R in Action: Data Analysis and Graphics with R

Kabacoff (2015), 2nd ed., Manning

Chapters 6 and 7.1–7.2

Data Visualization with R

Kabacoff (2019) · rkabacoff.github.io/datavis

Chapter 2

Basics of ggplot2 for Beginners

Padmanabhan (2019)

rpubs.com/arvindpdmn/ggplot2-basics